Azure Secure Enterprise Project

Architecture, Deployment Walkthrough, and Service Rationale

Author	Mina Gaballa
Primary Scope	Secure Azure administration, networking, traffic control, monitoring, and private service access
Regions	Norway East, Central India, Canada Central

This report documents the Azure lab build from architecture planning through validation, and explains the purpose of each major Azure service that was selected in the solution.

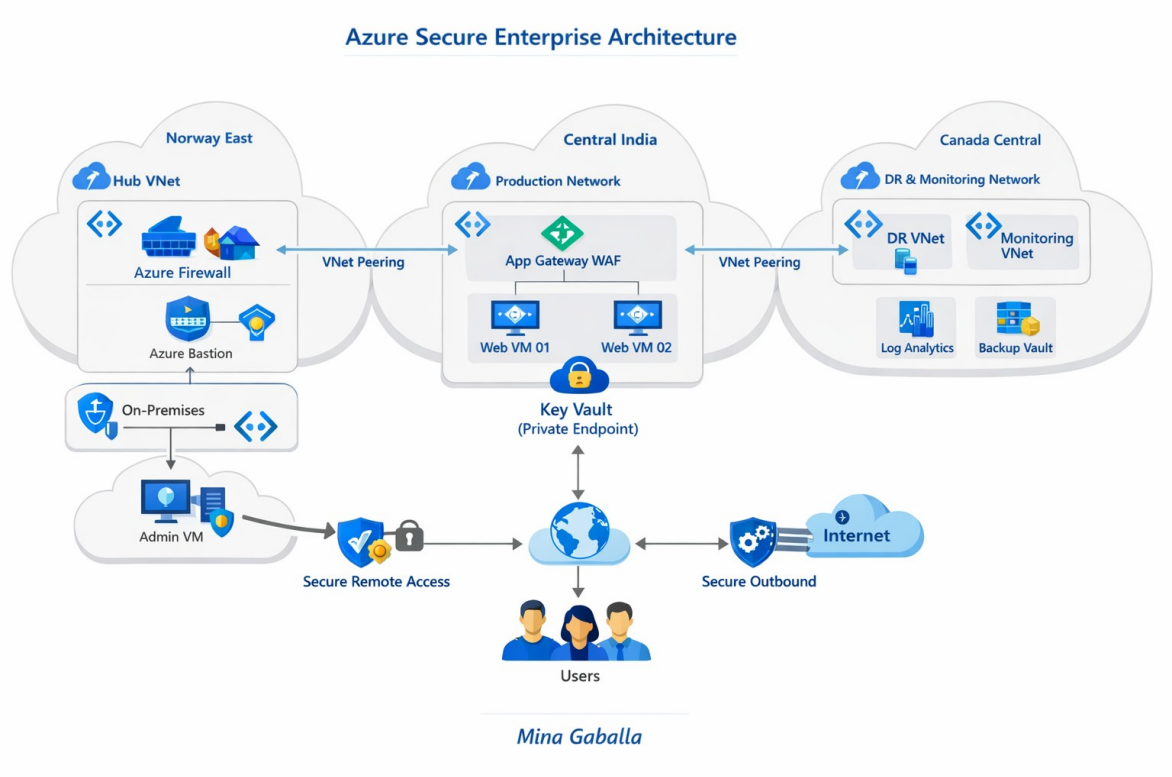

Figure 1. High-level architecture used for the Azure Secure Enterprise Project.

1. Project Overview

The project was built to simulate a secure enterprise-style Azure deployment that combines infrastructure administration, network design, and cloud security controls in one practical environment. The implementation aligns with the kinds of tasks covered by AZ-104, AZ-700, and AZ-500 by joining management access, protected application delivery, centralized outbound filtering, monitoring, and private connectivity to platform services.

Project goals

- Build a hub-and-spoke Azure network that separates management, shared security services, and production workloads.
- Protect administrative access with Azure Bastion instead of exposing public RDP to virtual machines.
- Publish the web workload through Application Gateway with WAF and keep backend servers private.
- Control outbound traffic from the production subnet by routing it through Azure Firewall.
- Use Key Vault with Private Endpoint and private DNS to keep sensitive platform access off the public internet.
- Collect monitoring data in Log Analytics and enable Microsoft Defender for Cloud for posture visibility.

Architecture summary

- Hub region: Norway East hosts Azure Firewall, Azure Bastion, and shared connectivity services.
- Primary workload region: Central India hosts Application Gateway, two IIS-based web servers, route control, and Key Vault private access.
- Additional region: Canada Central extends the design and demonstrates multi-region network planning.
- Connectivity pattern: global VNet peering connects the hub to spoke networks while preserving a clean operational boundary between services.

2. Azure Services Used and Why They Were Selected

The table below explains each major Azure service, what it does in the project, and why it was chosen over a less controlled alternative.

Service	Category	Role in the project	Why it was used
Resource Groups	Administration	Separate management, hub, production, and monitoring resources logically.	Improves organization, lifecycle control, and role assignment boundaries.
Virtual Networks	Networking	Provide isolated address spaces for hub and workload regions.	Foundation for segmentation, routing, and private service placement.
Global VNet Peering	Networking	Connect hub and spokes across regions.	Enables low-friction private connectivity without public exposure.
Azure Bastion	Security / Access	Provide browser-based secure RDP to VMs.	Avoids public IP-based administrative access and reduces exposure.
Admin VM	Management	Acts as the management host for validation and administration.	Provides a practical operations endpoint inside the environment.
Web VMs	Compute	Host the IIS-based application workload.	Creates a realistic backend tier for load balancing and health validation.
Application	App Delivery	Publishes the web workload and	Combines Layer 7 load

Gateway + WAF		inspects HTTP/HTTPS traffic.	balancing with web application protection.
Network Security Group	Security	Protects the web subnet with controlled inbound rules.	Adds subnet-level enforcement and least-privilege network access.
Azure Firewall	Security / Networking	Inspects and controls outbound traffic from the workload subnet.	Centralizes egress filtering and documents controlled internet access.
Route Table (UDR)	Networking	Forces workload subnet traffic to use Azure Firewall as next hop.	Makes firewall inspection deterministic instead of optional.
Key Vault	Security / Platform	Stores sensitive values and demonstrates private service access.	Represents secure secret management in a production-style design.
Private Endpoint + Private DNS	Networking / Security	Provides private-only access to Key Vault from the workload network.	Prevents public access and keeps service resolution internal.
Log Analytics Workspace	Monitoring	Collects diagnostics from Application Gateway and Firewall.	Creates a single place for operational and security review.
Defender for Cloud	Security Posture	Adds security recommendations and subscription-level posture visibility.	Shows how cloud resources are reviewed against security best practice.

3. Deployment Walkthrough

The environment was built in a deliberate order so that dependencies were available before security and application components were added.

1. **Create resource groups:** Separate resource groups were created for hub, management, production, and monitoring to keep the solution easy to manage and document.
2. **Deploy virtual networks:** A hub VNet was created in Norway East, a workload VNet in Central India, and an additional VNet in Canada Central.
3. **Add subnets:** Dedicated subnets were defined for AzureFirewallSubnet, AzureBastionSubnet, management, web, private endpoints, and Application Gateway.
4. **Configure global VNet peering:** Hub-to-spoke peering was configured to connect the regions while maintaining a central hub model.
5. **Deploy Azure Bastion:** Bastion was placed in the hub so management traffic could stay private and browser-based.
6. **Deploy management VM:** A Windows administrative VM was deployed without a public IP and was accessed through Bastion.
7. **Deploy web VMs:** Two Windows web servers were deployed in Central India to represent the workload backend.
8. **Install IIS and prepare validation pages:** IIS was installed on both backend VMs and each VM served a unique page to prove traffic distribution.
9. **Deploy Application Gateway with WAF:** Application Gateway was placed in a dedicated subnet, configured with a public frontend, backend pool, HTTP settings, listener, rule, and backend health validation.

10. **Apply web subnet NSG:** The web subnet NSG allowed HTTP from the Application Gateway subnet and supported controlled administrative access.
11. **Deploy Azure Firewall:** Azure Firewall was placed in the hub to inspect and control workload egress.
12. **Create and associate route table:** A route table was attached to the web subnet and set the firewall private IP as the next hop for 0.0.0.0/0.
13. **Create firewall application rules:** Rules were added to allow only selected outbound destinations for validation, demonstrating controlled internet access.
14. **Enable diagnostics and monitoring:** Log Analytics diagnostic settings were enabled for Firewall and Application Gateway.
15. **Enable Defender for Cloud:** Defender for Servers was enabled to demonstrate subscription-level security posture visibility.
16. **Deploy Key Vault with Private Endpoint:** Key Vault public access was disabled and a private endpoint with private DNS was configured to prove private service consumption.

4. Validation Evidence

The following screenshots capture the main implementation checkpoints and confirm that the solution behaved as designed.

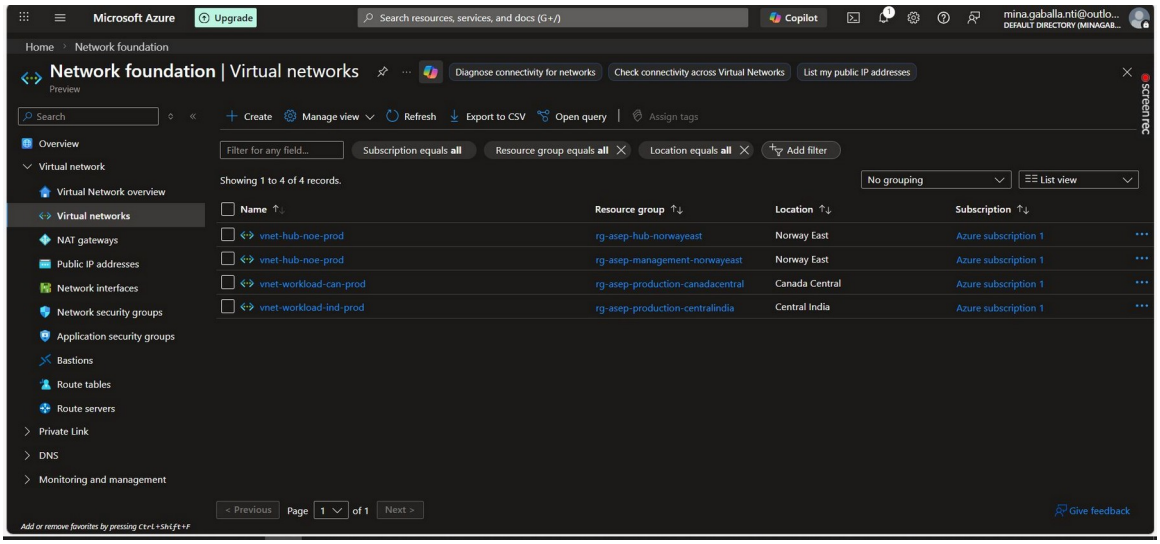

Figure 2. Virtual networks deployed across the selected Azure regions.

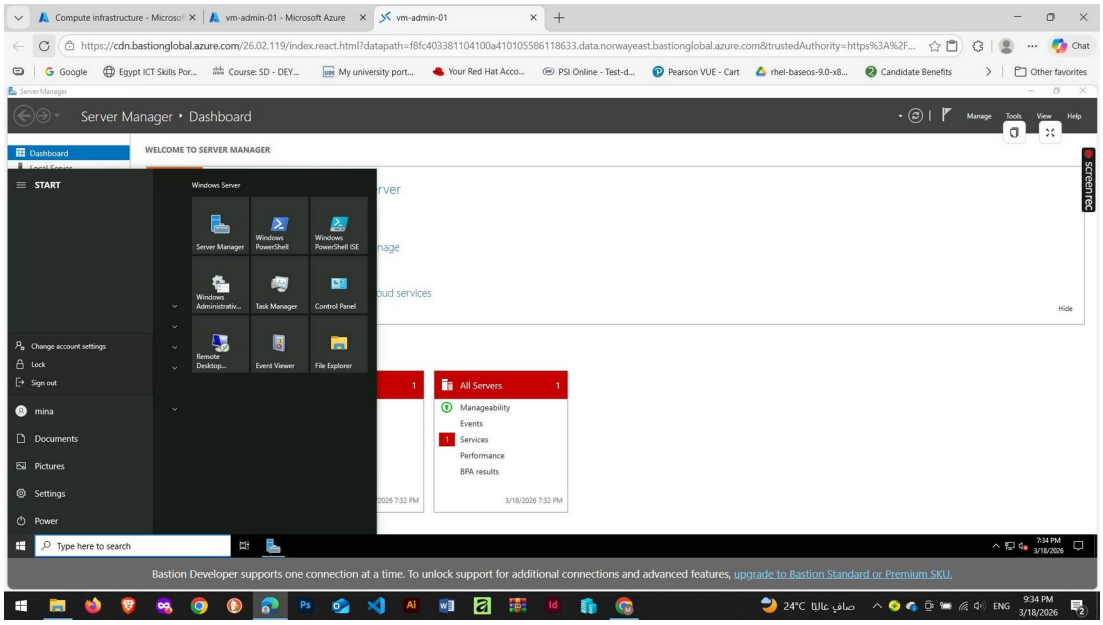

Figure 3. Secure administrative access to the management VM through Azure Bastion.

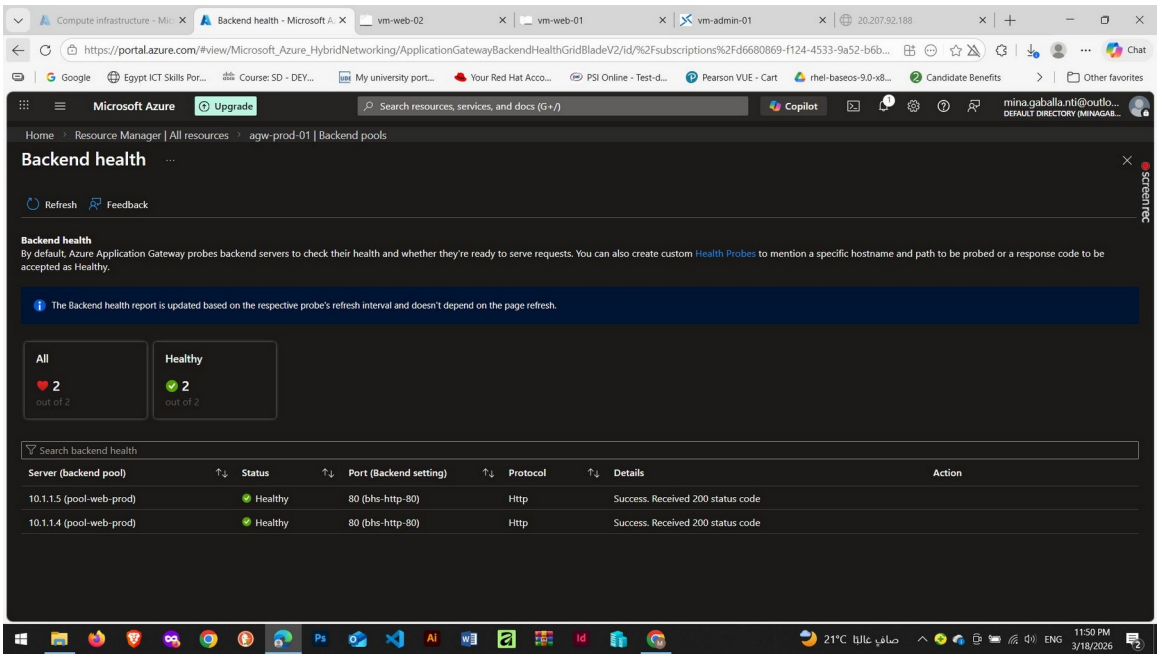

Figure 4. Healthy backend status for the web servers behind Application Gateway.

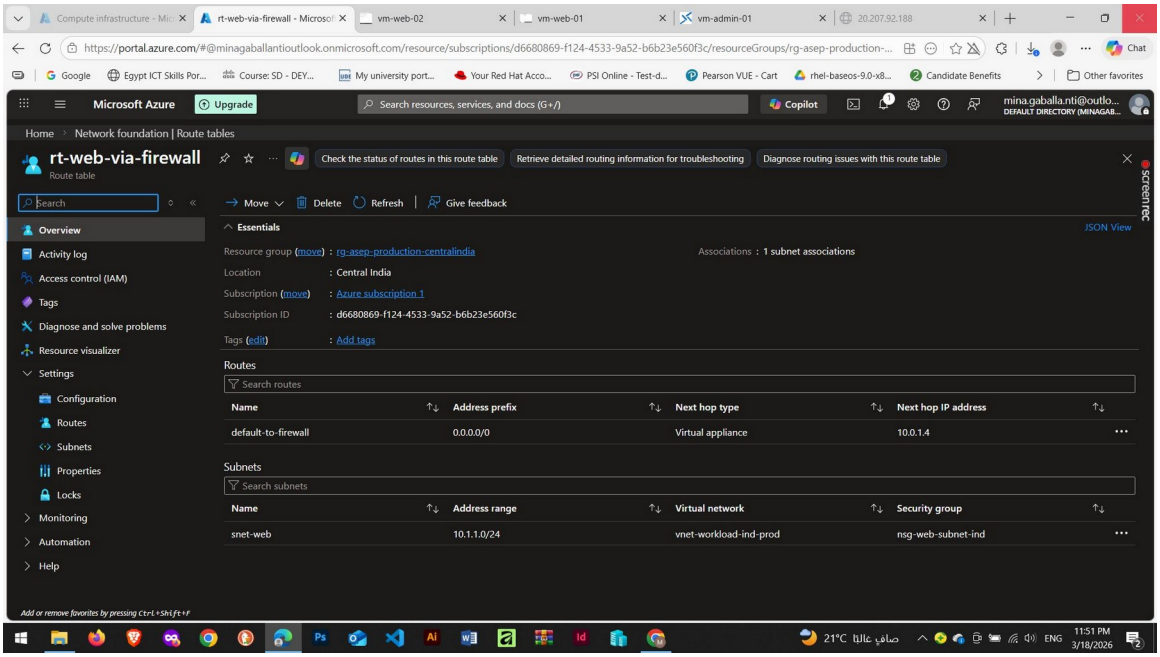

Figure 5. Route table used to force workload subnet egress through Azure Firewall.

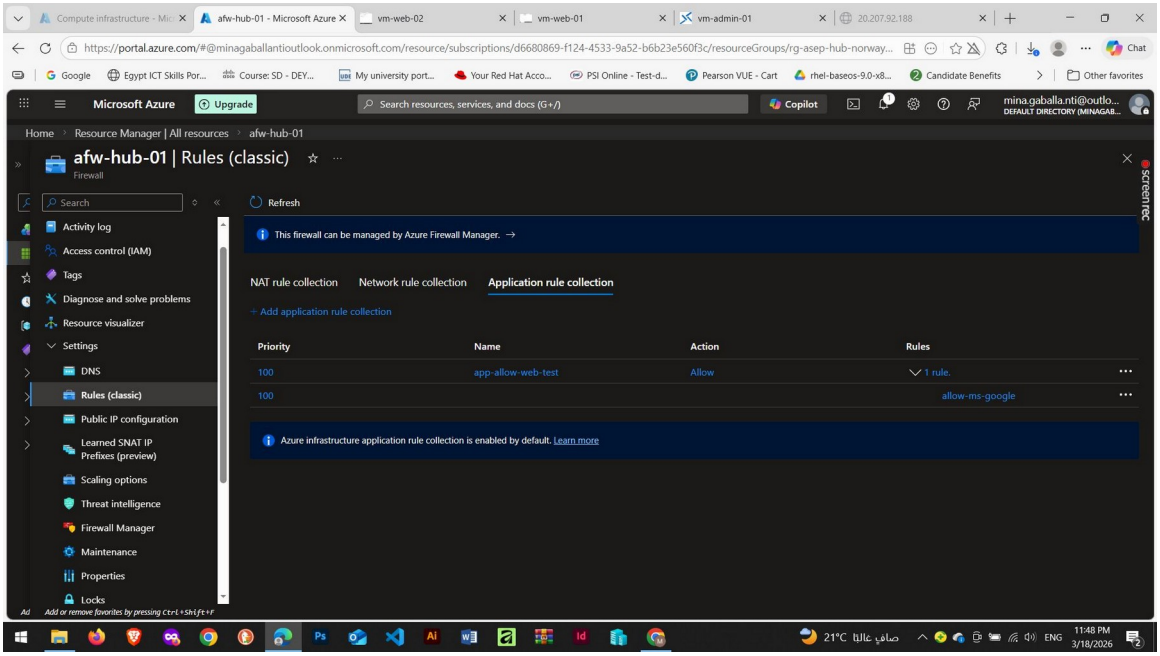

Figure 6. Firewall application rule collection used to control outbound web access.

5. Security Design Summary

- No public RDP exposure was required for virtual machines because Azure Bastion handled administrative access.
- The web workload remained private behind Application Gateway, while backend health checks validated operational readiness.
- The workload subnet used a route table to ensure outbound traffic was inspected by Azure Firewall.
- Subnet-level control was enforced through an NSG rather than relying only on default platform behavior.
- Key Vault public access was disabled, and the service was consumed through a private endpoint with private DNS integration.

- Diagnostics from the Firewall and Application Gateway were directed to Log Analytics to support operational review.
- Defender for Cloud provided security posture visibility and highlighted how the subscription could be reviewed against security recommendations.

6. Key Learning Outcomes

- How to structure a multi-region Azure lab with a hub-and-spoke design that remains easy to explain and document.
- How Azure Bastion, Application Gateway, Azure Firewall, NSGs, and UDRs combine to create layered network protection.
- How private endpoints and private DNS change service resolution behavior and support private-only platform access.
- How to validate an Azure design using backend health, route association, firewall rules, DNS resolution, and screenshots.
- How to present a hands-on cloud project professionally for GitHub, portfolio review, and interview discussion.

7. Referenced Deployment Artifacts

The project was also documented with PowerShell and JSON files that matched the build sequence used during implementation.

- 01-create-resource-groups.ps1
- 02-create-virtual-networks.ps1
- 03-add-subnets.ps1
- 04-create-global-vnet-peering.ps1
- 05-create-vm-admin-01.ps1
- 06-create-vm-web-01.ps1
- 07-create-vm-web-02.ps1
- 08-install-iis-on-web-servers.ps1
- 09-create-route-table-via-firewall.ps1
- 00-project-variables.json
- 10-portal-created-resources.json

8. Conclusion

This Azure Secure Enterprise Project demonstrates a complete practical workflow: architecture planning, controlled deployment order, layered security implementation, workload validation, and portfolio-quality documentation. The result is a strong cloud project that can be discussed from both an implementation perspective and a design rationale perspective.

Mina Gaballa